Enhancing Postural Monitoring in Wheelchair Users Through Context Classification

Nerea Perez, Aitziber Mancisidor, Itziar Cabanes, *Member, IEEE*, and Patrick Vermander

***Abstract*— Globally, the number of wheelchair users is steadily increasing. These people often adopt sitting patterns that reflect their functional status. Monitoring the user's postural status can help users and healthcare professionals to treat them. However, this posture is sometimes influenced by the environment in which the chairs move, and not necessarily by changes in their functional status. To address this problem, this study presents a model designed to classify wheelchair movement contexts, enabling the identification of what is happening in the user's environment. To do this, data has been collected using a robust and non-intrusive combined monitoring system, which records both the wheelchair's movement and the user's posture. These data have been used to train classifier models capable of distinguishing between seven categories of environments that are common in the daily lives of wheelchair users: flat surface, ramp up, ramp down, right turn, left turn, obstacles, and abrupt braking. These models have been developed using machine learning techniques, such as K-Nearest Neighbors (KNN), Artificial Neural Networks (ANN) and Support Vector Machines (SVM). The results show an accuracy of 90% in free-running tests and more than 99% in controlled runs. These results remained consistent despite variations in training subjects, validated by leave 2 out cross-validation. This innovative approach has the potential to improve the quality of life of wheelchair users by providing an accurate and effective tool to understand and address complex interactions between the environment and the users' posture.**

***Index Terms*— Biomedical monitoring, wheelchairs, smart healthcare, context modeling, intelligent sensors.**

I. Introduction

GLOBALLY, according to the World Health Organization (WHO) there are 80 million people who depend on the use of wheelchairs for mobility [1]. Users usually have mobility limitations, often associated with medical-related conditions. They typically adopt habitual sitting postures, which may reflect their functional status [2]. Observation, monitoring and analysis of these postural patterns prove to be crucial elements in understanding the overall health status of a person with reduced mobility. This information not only provides a detailed view of the individual's physical condition but also enables health professionals to tailor and provide more accurate and effective treatment.

In this context, numerous studies have explored the development of postural monitoring devices to improve medical care [3], [4], [5], [6], [7], [8] and the development of intelligent systems to detect different postural anomalies [9]. Despite advances in postural monitoring, healthcare professionals and wheelchair users state that physical factors such as ramps, turns, and braking greatly affect posture [10]. This influence can compromise the user's posture even without a change in their functional status, leading to misinterpretations. Therefore, it is essential to monitor wheelchair movements to maintain constant awareness of the environment in which the wheelchair is moving and thus determine the origin of the postural change. Therefore, in order to be able to carry out a continuous evaluation, the development of an intelligent system capable of interpreting data and assessing the wheelchair's environment is crucial.

Several studies have focused on monitoring the wheelchair environment for various purposes, including autonomous driving [11], [12], accessible path finding [13], [14], fall detection [15], [16], and measuring wheelchair propulsion patterns [17], [18]. These applications exemplify the integration of machine learning into wheelchair navigation, offering more efficient and accessible solutions for people with mobility disabilities. However, these developments often overlook the improvement of users' health status. They are mainly focused on providing assistance to facilitate daily life, rather than addressing medical aspects. Most of these solutions do not combine context information with postural diagnosis or rehabilitation. While some studies monitor physiological variables to identify risks [19], the information collected is often insufficient to assess the patient's functional status and assist in rehabilitation with physiotherapists. Although these studies have contributed to improving users' living comfort, they have failed to comprehensively address the relationship between the environment, posture, and rehabilitation. This lack of integration limits the development of more individualized and effective solutions to significantly improve the quality of life of wheelchair users.

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The authors are with the Department of Automatic Control and System Engineering, Bilbao School of Engineering, University of the Basque Country-UPV/EHU, 48013 Bilbao, Spain (e-mail: aitziber.mancisidor@ehu.eus).

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Therefore, the development of an intelligent system capable of processing and classifying the data obtained from the wheelchair's environment is essential to have an effective control of the surroundings. This classifier will allow to identify and categorize different elements present in the environment, such as obstacles, slopes or potholes. Furthermore, to maximize the usefulness of this information and to correlate it with the user's functional status, it is necessary to simultaneously collect data on the user's posture. This makes it possible to obtain a valuable dataset to apply machine learning techniques and provide quantifiable results to healthcare personnel.

Regardless of the application, the sensors employed for environment or context monitoring often share similarities. Among these are vision sensors, such as LIDAR cameras [12], [20], which are prevalent due to their ability to offer comprehensive visual data [16], [21]. However, despite their utility, vision sensors can be costly and may not provide as quantifiable information as other devices. Wearable sensors, like accelerometers attached to individuals [22], [23], present another popular option. These are commonly utilized for monitoring physiological parameters or movement patterns, which, while related to the environment, do not directly identify it. Additionally, when worn by the user, they can be intrusive and prone to placement errors. Therefore, for environmental monitoring purposes, utilizing IMUs or accelerometers directly on the wheelchair provides more accurate results [24], [25], [26], as these devices directly measure wheelchair accelerations, angular velocities, and orientations. To augment the data collection process, in addition to utilizing accelerometry, distance traveled can be tracked using the global positioning system (GPS) [12], [15]. Nevertheless, a limitation of this system is its inability for indoor tracking. To address this, there are simpler alternatives, such as calculating distances traveled using encoders, as well as more intricate solutions involving QR code information points coupled with a camera and autonomous navigation systems [24].

Moreover, it is difficult to find studies that apply classification techniques and obtain clear categories for each wheelchair environment. Some studies employ the classification of various terrain types for the detection of pavement irregularities [27] or for the development of autonomous driving systems [28], [29]. Furthermore, although the primary scope is the classification of accessible routes [14], [30] and to avoid obstacles or falls that may affect wheelchair movement [31], [32], none of these studies classify the context through which the wheelchair moves and establish defined categories for each of them. If they do, it is usually in a simulated environment and with a limited number of classes [33].

The purpose of this paper is to present an intelligent classifier designed to classify the environments in which a wheelchair moves, in order to enrich postural monitoring of the user and help healthcare professionals to treat them. To this end, data have been collected using a robust and non-intrusive combined monitoring system capable of recording both wheelchair movement and user posture. These data have been compared with three most common classification techniques: KNN, SVM and ANN, in order to develop a generalized environment classifier. This classifier is able to differentiate between seven types of most common day-to-day environments for wheelchair users: ramp up, ramp down, left turn, right turn, braking, potholed surfaces and flat land. To train it, data collected from different subjects were used and a leave-2-out validation was implemented to obtain a generalized classifier. In addition, the robustness of these model has been validated by random runs through urban environments.

This integrated approach not only seeks to classify the environments in which the wheelchair user navigates, but also to understand how these environments impact their posture. Since posture is a reflection of a person's functional status, this classification system not only offers a tool to adapt to different contexts, but also provides valuable information for physical therapists. This allows professionals to develop more specific and effective interventions to improve the health and quality of life of those who depend on wheelchairs.

The rest of this article is structured as follows. Section II describes the monitoring system and the database collection process. Section III outlines the methodology for training the appropriate classifier and discusses the test results. Section IV elaborates on the validation trials and provides a discussion. Finally, in Section V, the main ideas are summarized, and future work is outlined.

II. Data Base Generation

In order to design a classifier of wheelchair user contexts, a suitable database is needed. With this purpose, this section briefly describes the combined monitoring device developed for data collection (Section II-A) and the experiments conducted to generate the database of different contexts (Section II-B).

A. Combined Monitoring System

The combined monitoring system developed by the authors integrates postural monitoring and real-time wheelchair movement data collection, addressing the critical need for accurate information on wheelchair users' environment [34].

The device is divided into two modules: one for monitoring wheelchair movements (green in Fig. 1) and the other for postural monitoring (purple in Fig. 1). The first module integrates an Xsens Inertial Measurement Unit (IMU) on the chassis, together with two encoders composed of inductive sensors placed on each wheel. Various parameters are obtained from these sensors, including three-axis linear accelerations (a_x, a_y, a_z), angular velocity (ω_x, ω_y, ω_z) and orientation (ϕ_x, θ_y, ψ_z) plus the distance traveled by each wheel (d_{wheel1}, d_{wheel2}).

The second module is the i-KuXin system, a self-developed postural monitoring system that uses 16 non-invasive Force-Sensing Resistor (FSR) sensors strategically placed on a portable cushion, which is positioned on the wheelchair. The force measurements of the i-KuXin system are provided for both the seat ($S_1 \dots S_8$) and the backrest ($B_9 \dots B_{16}$). This portable two module system results in a total of 27 variables specified in Table I.

The designed sensor module and added sensors are robust, accurate, and cost-effective, making them suitable for

TABLE I
SUMMARY OF THE MEASURED VARIABLES AND USED SENSOR TYPES

Variable	n	Unit	Sensor type	Symbol
Acceleration	3	m/s^2	IMU	a_x, a_y, a_z
Angular velocity	3	rad/s	IMU	ω_x, ω_y, ω_z
Orientation	3	degrees	IMU	ϕ_x, θ_y, ψ_z
Distance	2	meters	Encoder	d_{wheel1}, d_{wheel2}
Seat Force	8	mNewton	i-KuXin (FSR)	S_1...S_8
Backrest Force	8	mNewton	i-KuXin (FSR)	B_9...B_{16}

Fig. 1. Placement and coordinates of sensors on the Sunrise Medical QUICKIE Q200 R wheelchair. Purple: i-KuXin and FSR sensors. Green: Xsens MTI-3-DK IMU. Blue: Inductive sensors.

TABLE II
SUMMARY OF TESTS CARRIED OUT FOR THE DATABASE GENERATION

Label	Class	Test	Velocity
0	Flat surface	Constant velocity	1-5
1	Ramp up	4º ramp	1-2
		7º ramp	1-2
2	Ramp down	4º ramp	1-2
		7º ramp	1-2
3	Potholes	Tram tracks	1-2
4	Turn right	90º turn	1-2
		180º turn	1-2
5	Turn left	90º turn	1-2
		180º turn	1-2
6	Braking	Discontinuous velocity	1-5

TABLE III
RELEVANT DATA OF THE PARTICIPANTS

Subject	Gender	Age	Weight (kg)	Height (cm)	BMI* (kg/m^2)
S1	M	31	77	180	23.8
S2	M	27	86	183	24.9
S3	M	29	64	185	18.7
S4	F	28	68	166	24.1
S5	F	26	67	170	23.2

Note: Body Mass Index (BMI)

long-term use and adaptable to various wheelchair models. Moreover, data acquisition is facilitated through different Arduino boards connected to the sensors, enabling wireless connectivity via Bluetooth for real-time transmission to remote devices. The system, powered by the wheelchair's battery, has an autonomy of over 24 hours, allowing continuous data collection with a sampling rate of 5 Hz. Its energy consumption is negligible, representing less than 0.1% of the wheelchair's total battery usage. The developed device, accessible through mobile and PC applications, provides a comprehensive solution for monitoring and optimizing the wheelchair user's experience. In this case, the wheelchair chosen for testing the system is the Sunrise Medical QUICKIE Q200 R, which offers five different driving speeds: 3 km/h, 5 km/h, 7 km/h, 9 km/h, and 11 km/h.

B. Conducted Test

In order to develop an environment classifier it is necessary to generate a complete database. For this purpose, a series of dynamic seating tests have been conducted to reproduce various scenarios and challenges encountered by wheelchair users in their daily lives. These scenarios have been identified based on recommendations from healthcare professionals and wheelchair users, highlighting locations and situations such as ramps, different pavement types, potholes and turns. To understand the impact of these environments or contexts, the authors conducted a prior correlation study between posture and wheelchair motion variables using the monitoring device mentioned in the previous paragraph [34].

The slope of the floor can influence the posture of the user causing the torso to lean, so ramps have therefore been evaluated. In addition, turns can alter the user's weight distribution and therefore posture, which in certain situations may increase the risk of falling. Potholes and braking can also affect the user's balance and, even on flat ground, speed can generate vibrations that affect the user's posture. Consequently, as summarized in Table II, seven categories or labels have been established: ascending ramps, descending ramps, right turns, left turns, encountering potholes, abrupt braking's, and navigating flat surfaces.

To identify these categories, supervised tests have been performed in various contexts. These included trials involving acceleration and deceleration tests in 5 different speeds in flat surfaces, ascending and descending ramps at inclinations of 4° and 7°, trials crossing train tracks or encountering potholes, 90° and 180° turns and braking. Table II shows a summary of the tests carried out and at what velocities. These tests aimed to simulate real-scenarios and challenges faced by wheelchair users, providing valuable data to train the classifier. These tests have been performed by five (N = 5) different subjects using the monitoring system and the wheelchair described in the previous section. Also, a more detailed description of the test subjects' physical characteristics is provided in Table III.

The test have been conducted under the authorization of the ethics committee of the University of the Basque Country (M10-2022-007). At the end, more than 100 tests (including all environments and velocities) have been obtained, generating almost 19,000 samples per subject. Each of these samples contains a total of 27 sensor values as mentioned above. These can be classified into two groups: dynamic variables and

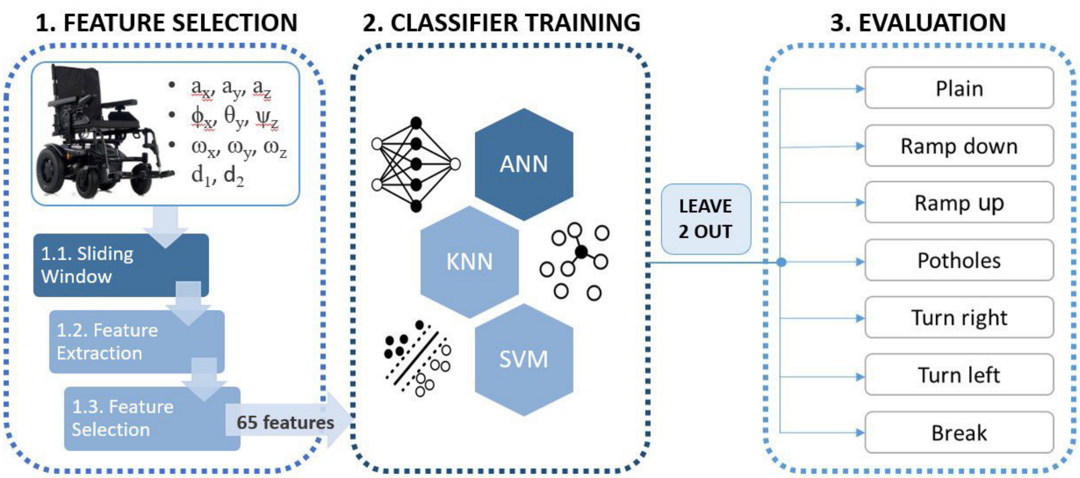

Fig. 2. Scheme of the methodology used for the context classifier, consisted into three steps: feature selection, classifier training and evaluation.

postural variables. As it is intended to classify the environment only the 11 dynamic variables will be used in this study.

III. Methodology

This section describes in detail the methodological approach used to design the context classifier of a wheelchair. The developed methodology is divided into three steps (Fig. 2). The creation of the sliding window and the data pre-processing will be explained for the feature selection (Section III-A), as well as the selection of techniques used in the study to train the classifier (Section III-B). Furthermore, the training process using leave 2 out cross validation to assess the robustness and effectiveness of the environment classification system will be discussed (Section III-C).

A. Feature Selection

The first step in this methodology proposed for the design of the classifier is feature selection (Step 1 in Fig. 2). Signals are filtered and normalised, though only minimal pre-processing is required due to sensor stability. To capture the dynamic character of the moving wheelchair, a sliding window approach is proposed (Step 1.1), as it enables to capture time variations and manage non-stationary data. Several sliding windows with various widths and overlaps have been evaluated. Window width refers to the number of data points included in each segment, while overlap represents the number of data points shared between consecutive windows. Finally, a width of 10 samples per window and an overlap of 7 data points have been selected (Fig. 3) as they give the best results for different environments and speeds.

In each sliding window, the 11 dynamic variables have been selected together with their respective 10 samples to calculate various features. These features are then fed into the classifier. A total of nine commonly used [35], [36], [37] statistical features have been extracted for each variable, including mean, standard deviation, variance, median, 25th and 75th percentiles, minimum, maximum and range. This results in a total of 99 features extracted for each sliding window (Step 1.2 in Fig. 2).

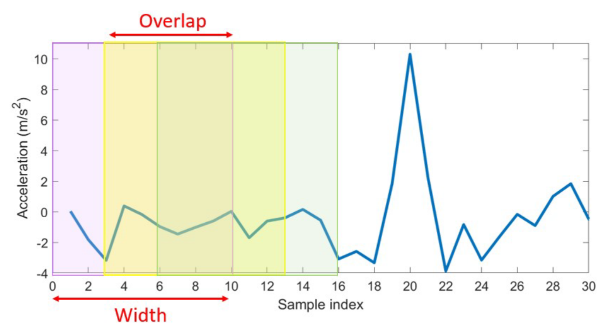

Fig. 3. The sliding window approach segments the data into windows of 10 samples, with an overlap of 7 samples between consecutive windows.

In order to explore the possibility of reducing the number of features, a detailed analysis has been carried out using methods such as Random Forest and Relief. In this analysis, specific parameters have been configured, setting the number of trees to 100 for Random Forest and the number of neighbors to 20 for Relief. From the analysis of the results, it was found using the two techniques that there were some parameters that were not relevant. Therefore, in the feature selection, it was decided to reduce the number of features to 65 (Step 1.3 in Fig. 2).

B. Context Classifier Training

After the generation of sliding windows and the selection of features, the classification techniques for training have been selected (Step 2 in Fig. 2). There are several training techniques available, among the most common supervised techniques are K-nearest neighbors (KNN), [29], [30], Support Vector Machines (SVM) [29], [30], [31], Artificial Neural Networks (ANN) [30] or other deep learning techniques [27], [30]. These classifiers were chosen for their ability to capture nonlinear patterns and complex interactions between sensor signals, which are often influenced by multiple simultaneous factors such as terrain type, speed, and user behavior. Unlike simpler models, these techniques allow for more accurate discrimination between overlapping or ambiguous situations.

After selecting the classification techniques, the training and test data sets have been prepared. As for each specific context a different number of velocities were tested, not all classes have the same number of windows, which can lead to bias in model training. To prevent this, the model has been trained using balanced data sets, ensuring that no specific class was favored.

To achieve this, in the total training set all windows have been initially grouped according to their respective classes. The class with the lowest representation was identified and 70% of that class was selected for training, leaving the remaining 30% for testing. Using this number as a reference, the same number of windows have been selected from the other classes. This process has been carried out to form a balanced training matrix. The remaining data has been reserved for the testing phase, thus ensuring that the model is evaluated with independent information.

Moreover, the optimal hyper-parameters of each technique have been tuned to guarantee optimal performance in the classification of the dynamic data of the wheelchair. In particular, for KNN, one neighbor has been found to be the most optimal, both in terms of accuracy and computational cost. On the other hand, in SVM, Bayesian Optimization techniques have been used to determine the optimal values of the hyper-parameters. For ANN, a multi-layer perceptron neural network, a detailed analysis focusing on the number of neurons in the hidden layer has been performed. This process has evaluated the accuracy and standard deviation associated with different neuron configurations resulting on 11 neurons in the hidden layer.

C. Leave 2 Out Cross Training and Evaluation

This classifier is focused on contributing to the postural diagnosis of users, recognizing that posture is a highly individualized aspect. Nonetheless, given that environmental variables often show similarities across different users, the generalization capability of the context classifier model has been evaluated. This aims to ensure that the classifier can adapt to diverse users while still providing accurate insights for diagnostic purposes.

Thus, the Leave 2 Out Cross Validation (LPOCV) technique has been implemented to evaluate the generalization capacity of the classification models (Step 3 in Fig. 2). This consists of dividing the database of n subjects into a training set composed of n-p subjects and a validation set with p subjects, iterating through all possible combinations of these sets. For this specific case, with data from 5 subjects available and the goal of selecting 3 for training and 2 for testing, 10 possible combinations are created.

Context classifiers of ANN, SVM and KNN have been trained using these 10 combinations, and satisfactory results have been obtained. To analyze the results, the accuracy and deviation of the resulting models are shown in Fig. 4.

By observing the accuracy bar graph, all possible combinations reach an accuracy higher than 92%. This consistency suggests that the choice of subject for training and validation has minimal impact on results, highlighting the classifier's ability to generalize across different subjects. Moreover, it is observed that the ANN-trained models show the best results, demonstrating lower variability in their predictions compared to other algorithms. This tendency is not unexpected, as neural networks tend to be better able to make robust generalizations due to their ability to learn complex patterns and adapt to various conditions.

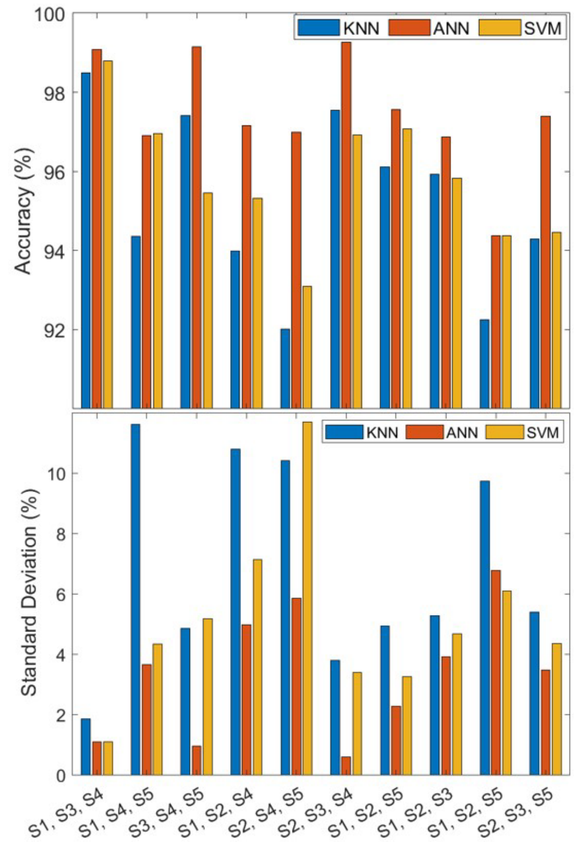

Fig. 4. Comparison of accuracy (top) and standard deviation (bottom) for KNN (blue), ANN (red), and SVM (yellow) models using leave-2-out cross-validation. The X-axis shows the participants used for training in each iteration, while the excluded ones were used for testing.

The second most accurate technique for classifying after ANNs proves to be SVM. However, it is important to mention that this technique entails a high computational cost in the training. On the other hand, the KNN algorithm shows the lowest accuracy, even though it exceeds 92%, reaching 98% in some cases.

When examining the deviations of the models, it was observed that ANN is also the algorithm with the lowest deviation, not exceeding 7 points, while SVM and KNN reached almost 12. In general, the developed classification models show high performance even when faced with different combinations of subjects in the training and test sets. This evidences the possibility of developing a generalized classifier for users with different physical characteristics.

After establishing the feasibility of a generalized approach, it has been decided to use data from all subjects to train the classifier, with the aim of developing a single, efficient model. A single database has been created with 5 subjects for the training and testing.

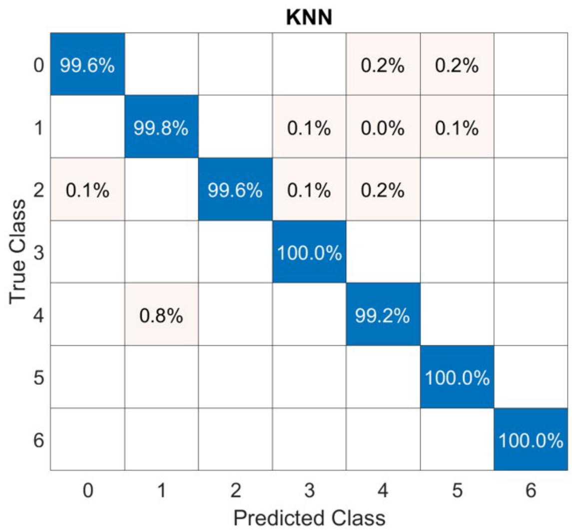

Fig. 5. Confusion matrix of the KNN model trained with data from all five subjects. The classes correspond to: 0 – plain, 1 – ramp up, 2 – ramp down, 3 – potholes, 4 – right turn, 5 – left turn, and 6 – braking.

Specifically, the KNN classifier shows excellent performance, with accuracy over 99% across all classes as shown in Fig. 5. However, there is some minor confusion between certain classes, primarily between Class 0 (Plain) and the turns (Classes 4 and 5), which may share similar characteristics in certain contexts. Despite this, the overall results remain highly reliable, with an average accuracy of 99.7%. Other techniques also performed well, with ANN achieving an average accuracy of 99.1% and SVM reaching 98.5%. It is important to underline that all the percentages are remarkably high. This fact demonstrates the ability of these algorithms to handle the overall database, highlighting the effectiveness of a more unified and generalized approach to data classification and validates the proposed methodology.

IV. Validation and Discussion

After designing the context classifier and validating that the proposed methodology is adequate in different environments and users, its correct functioning has been validated in free trials. As a context classifier, the objective of this model is to categorize any surrounding environment that the wheelchair traverses into one of the predefined context categories. In this sense, validation tests are designed and described in Section IV-A, and a detailed results and discussion of the final findings is given in Section IV-B.

A. Validation Test Design

Free walking tests have been carried out to validate the effectiveness of the classification models in a more realistic environment. Two subjects, V1 and V2, participated (their characteristics are detailed in the Table IV) in these runs, which lasted approximately 20 minutes each, generating nearly 6000 data samples. The participants moved freely through urban areas with no predefined route, facing real-world mobility conditions including crowds and unpredictable obstacles, particularly during midday hours when pedestrian traffic was high.

TABLE IV
Relevant Data of the Validation Tests Participants

Subject	Gender	Age	Weight (kg)	Height (cm)	BMI (kg/m^2)
V1	M	29	64	185	18.7
V2	F	28	68	166	24.1

During the routes, various obstacles and common situations have been encountered in the urban environment, such as ascending and descending ramps with inclinations of 2, 4 and 6 degrees, traffic lights or pedestrians who have forced the wheelchair to stop and brake, train tracks, crosswalks and potholes in the road. These environments have fit into the 7 context categories defined in the study, although there may be situations where several items occur simultaneously. Such conditions introduced uncertainty, requiring the classifier to robustly identify the predominant context despite overlapping or transient events.

Classifier performance was evaluated using manual labeling by researchers based on direct observation during the runs, classifying the evidence obtained in the seven predefined classes. Afterwards, the designed methodology has been applied using the collected data. First, data pre-processing has been performed, using a sliding window to extract the features of each window. These features have served as input for the previously developed classification models, which included the KNN, ANN and SVM algorithms.

B. Results and Discussion

Finally, with the new database collected for validation, the trained models have been evaluated. These models have been trained with all the subjects from the previous database. In Fig. 6, the results obtained by subject V1 during the free run are presented. In it, the accuracy of each model (ANN in red, SVM in yellow and KNN in blue) for each of the defined categories is shown. It can be seen that SVM achieves the best results, followed by ANN and finally KNN, with an average accuracy of 90.02%, 86.80% and 85.53% respectively.

In addition, Table V shows the hit percentages for each category. For example, ramp ups and downs show an accuracy over 93%. Furthermore, potholes are identified with an accuracy of 88%, and braking actions are recognized with around 85% accuracy. In contrast, turns and movements on flat terrain yield lower accuracy rates, close to 82%.

This variation in accuracy can be explained by the nature of the movements in each category. Ramps and potholes produce distinct sensor signals, making them easier to recognize. However, turns and movements on flat terrain share similar characteristics and therefore tend to be confused with each other. For example, on a typical route, it is difficult to distinguish between a turn and following a slightly curved road, as both generate similar data patterns. Similarly, what appears to be a straight line travel may include slight turns, which contributes to confusion in classification.

These results support the general performance of the proposed classification system, which represents a significant advance in the field of medical care. With accuracy rates exceeding 85% in free outdoor conditions and over 99% in

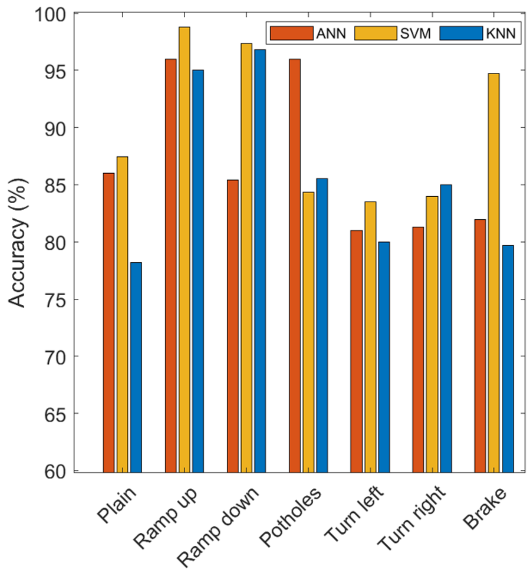

Fig. 6. Results of validation test of Subject V1 during free run.

TABLE V
PERCENTAGE OF ACCURACY BY TECHNIQUE AND CLASS

Class	Flat surface	Ramp up	Ramp down	Potholes	Turn right	Turn left	Braking	Mean
ANN	86,01	96,00	85,39	96,00	80,99	81,30	81,94	86,80
SVM	87,47	98,77	97,37	84,30	83,48	84,00	94,78	90,02
KNN	78,18	95,00	96,84	85,56	80,03	84,99	79,66	85,75
Mean	83,89	96,59	93,20	88,62	81,50	83,43	85,46	87,53

controlled scenarios, the methodology has proven to be both reliable and robust. Although SVM consistently outperformed the other models in terms of accuracy, the optimal choice will ultimately depend on the target application. In scenarios requiring real-time processing or constrained hardware, lighter models such as KNN or ANN may offer a more practical alternative. Overall, the system has demonstrated its adaptability and effectiveness under realistic and dynamic conditions.

A highlight of this work is the generalizability of the classification models. Through leave-2-out cross validation, it has been shown that the models maintain consistent and reliable performance regardless of the training subjects. This generalization ability is critical to ensure the effectiveness of the system in a variety of contexts, situations and users with different physical features.

Combining the results from the context classifier with the postural measurements from the sensors, this system provides physical therapists with an invaluable tool. It helps them better understand the impact of the environment on the posture and functional status of wheelchair users. Allowing the determination whether the postural change is due to the environment or to the functional state of the user. This information can be fundamental to the development of personalized treatment strategies and the optimization of patients' quality of life.

In summary, the study has demonstrated the feasibility and efficacy of a wheelchair environment classification system that provides a comprehensive and accurate assessment of the user's surroundings. This system has the potential to transform the way medical care for people with reduced mobility is approached, offering new opportunities to improve the life and autonomy of wheelchair users.

V. CONCLUSION

People with reduced mobility often have habitual sitting postural patterns, which reflect their functional status. Observation, monitoring and analysis of these patterns are essential to understand their overall health and to provide more accurate and effective treatment.

The aim of this study is to address this gap by presenting an accurate model for classifying the environments in which a wheelchair moves. It focuses on combining context recognition with postural monitoring of the user. To achieve this, data has been collected using a robust and non-intrusive combined monitoring system, capable of recording both the wheelchair's movement and the user's posture.

By collecting data from five subjects in seven different environments, a comprehensive database has been created and used to train a 7-environment classifier. The use of a sliding window and relevant feature extraction enables effective capture of temporal data. As a result, in validation tests, the classifier based on the SVM algorithm has shown an accuracy of 90% on free runs, which underlines its effectiveness in classifying dynamic environments.

Furthermore, the classifier has demonstrated a strong capacity for generalization, suggesting that the trained system can be applied effectively across different users, despite the highly individualized nature of postural assessments. Also, with this added contextual information about the wheelchair user's environment, physiotherapists can gain a deeper understanding of the postural data, going beyond isolated measurements to consider the influence of external factors on posture. This enriched data allows for a more comprehensive assessment of each patient's unique needs and conditions, potentially leading to more accurate diagnoses and more effective, personalized treatment plans.

Future work will focus on validating the system in clinical environments with users presenting diverse pathologies. Although the system is designed for easy adaptation, it still requires hardware adjustments to fit custom cushions (e.g., viscoelastic, air-based) and different wheelchair models. Additionally, efforts will be made to ensure smooth integration into low-power embedded platforms, facilitating real-time performance.

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